MUTUS v1.0: ODD Protocol

Item	Description
Model	MUTUS: Mutualism in Texts and Understanding of Students
Platform	NetLogo 7.0.4, as recorded in the supplied model; analysis in R (`MUTUS_analysis.R`)
Version	1.0 (model ladder M0–M5a)
Date	2026-10-26
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Parent model	STiMUS v2.1 (Appendix A)

The description follows the ODD protocol in its second update (Grimm et al. 2020). Sections 1–7 form the core ODD. Section 8 lists the simulation experiments, section 9 the analysis pipeline, and section 10 the main results and limitations; they are not part of the core protocol.

1. Purpose and patterns

1.1 Purpose

MUTUS is a minimal agent-based model of a **mutualistic network between two guilds.** One computational core supports two interpretations:

Aspect	Meadow world	Wiki world
Foragers	Pollinators (bumblebees)	Students of a course (e.g. psycholinguistics)
Resources	Flowering plants	Articles of a course wiki or glossary
Clusters	Plant species / patches	Topics of the course
Space	Landscape	Semantic space (distance = topical dissimilarity)
Trait matching	Handling ability, including a proboscis-length analogy, vs. corolla depth	Competence vs. conceptual difficulty (ZPD)
Service	Pollination	Editing, linking, improving the text

The model has three purposes:

1. **Theoretical.** To test which regularities of plant–pollinator networks (nestedness, modularity, persistence under adaptive foraging) transfer to collaborative text production, and which are changed by processes that are slow or absent on ecological time scales: learning within the zone of proximal

development (ZPD), scaffolding by text quality, and authoring of new texts by the learners themselves.

2. **Methodological.** To offer a generative mechanism for the mutualistic structure of socio-semantic networks, which has so far been described structurally (St-Onge et al. 2022).
3. **Pedagogical.** To serve as a learning object for non-STEM students: they explore the meadow first, then build and criticise the analogical mapping to the wiki. The mapping itself is a psycholinguistic object of study.

1.2 Patterns

Pattern	Source	Used for
P1. Nestedness above a null model that preserves degrees	Bascompte et al. 2003; Almeida-Neto et al. 2008	Structural realism, both worlds
P2. Moderate modularity aligned with species / topics	Olesen et al. 2007; St-Onge et al. 2022	Structural realism, both worlds
P3. Adaptive foraging raises persistence and reduces failed visits	Valdovinos et al. 2013, 2016	Mechanism check (M3)
P4. Flower-handling skill improves with experience and saturates	Laverty 1994; Chittka et al. 1999	Plausibility of M4 in the meadow

1.3 Hypotheses (tested in E1–E5)

- **H1 Neutral baseline.** Heterogeneous activity alone (M0) produces part of the network structure; trait matching and mutualistic feedback change it.
- **H2 Trait matching.** The ZPD band produces modular networks and failed visits; feedback (M2) and adaptive foraging (M3) reduce failures.
- **H7 Cross-cluster transfer.** Mixing topics is costly in the meadow (pollen dilution) and almost neutral in the wiki (interdisciplinary linking).
- **H7r Perception radius.** A narrow perceptual (topical) horizon suppresses nestedness and, in the wiki, exhausts novelty and causes dropout.
- **H8 Developmental rewiring (M4).** Learning shifts the ZPD upwards: learners outgrow resources (more too-easy visits) and rewire towards more complex ones. The original prediction that nestedness rises during a semester is tested here.
- **H9 Renewal (M5a).** When faded resources are replaced, the newcomer's complexity is set by its parent plant in the meadow and by its author in the wiki. Only in the wiki do resources therefore follow the learners.

2. Entities, state variables and scales

2.1 Entities

Entity	Breed	Default number	Notes
Forager	`foragers`	30 (`num-foragers`)	Bee / student
Resource	`resources`	98 (`num-resources`)	Plant / article; `who` = 0 … N−1 = `res-index`
Cluster	attribute `cluster-id`	6 (`cluster-count`)	`cluster-id = res-index mod K`
Space	`patches`	41 × 41, no wrapping	Background only

No links are created at run time. The bipartite network is accumulated in the foragers' `succ-visits` lists and exported as strings (§9).

2.2 Resource variables

Variable	Range	Meaning (meadow / wiki)
`complexity` c_k	[0, 1]	Corolla depth / conceptual difficulty
`eff-complexity` c_k^{eff}	[0, 1]	$c_k(1 - \sigma q_k)$; equals c_k in the meadow ($\sigma = 0$)
`reward` R_k	[0, 1]	Nectar standing crop / open contribution opportunities
`vitality` v_k	[0, 1]	Floral display / currency of the article in course discourse
`quality` q_k	[0, 1]	Pollination success / content quality
`alive?`	bool	$v_k \geq v_{\min} = 0.1$
`visit-count`, `service-count`	≥ 0	Visits and services since initialisation or the most recent renewal
`birth-tick`, `generation`	≥ 0	Tick and number of renewals of the slot (M5a)

2.3 Forager variables

Variable	Range	Meaning (meadow / wiki)
`activity` a_i	(0, 1)	Probability of acting in a tick
`energy` e_i	[0, 1]	Energy reserve / motivation
`skill` s_i	[0, 1]	Handling ability (proboscis length + learned handling) / competence
`skill-0`	[0, 1]	Skill at setup
`effort`	list, Σ = 1	Allocation of foraging effort / attention over resources
`familiarity`	list of counts	Visits to each resource (novelty in the wiki)
`succ-visits`	list of counts	Successful visits to each resource (network)
`mean-gain`	≥ 0	Running mean of gains

Variable	Range	Meaning (meadow / wiki)
`prev-resource`	resource / nobody	Source of carried pollen / concepts
`active?`	bool	Foraging / dormant (dropped out)
`cheater?`	false	Reserved field; no cheater mechanism is active in v1.0

2.4 Scales

One tick is one **opportunity to act**: a bee may or may not visit a flower, a student may or may not read or edit an article. A run is one flowering season or one semester of **150 ticks** (fixed). Space has no units; distance stands for landscape distance or topical dissimilarity, and the world does not wrap.

Both guilds are fixed sets of individuals: there is no reproduction of foragers. Loss is a state change (dormant forager, faded resource). From M5a on, a faded resource slot can be taken by a new resource (§7.10).

3. Process overview and scheduling

Each tick:

1. **Foragers** (random order). A dormant forager recovers (§7.8). An active forager forages with probability a_i (§7.1–7.7). It then pays the basal cost and becomes dormant if $e_i < e_{\min}$. A forager that was dormant at the beginning of the tick only rests, even if it reactivates during that tick.
2. **Resources.** Reward regeneration, decay of vitality and quality, update of effective complexity and of `alive?` (§7.9). Then renewal of faded resources (M5a, §7.10).
3. **Observation.** Update of monitors (Observation in §4).
4. **Tick.** The run stops at tick 150.

Within a visit the order is: consumption → outcome (fail / consume / service) → feedback to the resource → energy → bookkeeping → learning (M4) → effort update (M3). Matching and novelty are evaluated before these updates. Service uses pre-consumption energy and pre-learning skill; learning therefore affects matching from the next visit onwards. Effective complexity is recalculated during the resource-update stage, not immediately after each service.

4. Design concepts

Basic principles

Mutualism is a closed positive-feedback loop:
service → vitality and quality → reward and visibility → forager energy → service.
The loop is bounded by saturation (Holling type II), by depletion of reward and by trait matching. In the wiki the matching band is Vygotsky's ZPD (Vygotsky 1978), and quality lowers the effective difficulty of a text (scaffolding).

Emergence

Network structure (connectance, nestedness, modularity), persistence of both guilds, specialisation of foragers and the drift of resource complexity emerge from individual visits.

Adaptation

From M3 on, foragers shift effort towards resources that gave above-average gain (§7.6).

Objectives

Implicit: foragers gain energy. There is no fitness function.

Learning

From M4 on, skill grows after successful visits inside the ZPD band (§7.7), in both worlds: flower-handling skill in the meadow, competence in the wiki.

Prediction

None.

Sensing

A forager perceives resources within `perception-radius`, their vitality, and its own past gains. It does not perceive reward before the visit.

Interaction

Between guilds: consumption and service. Within guilds: indirect competition for reward (foragers) and for forager effort (resources). There is no direct forager–forager interaction and no stigmergy.

Stochasticity

Initial traits and positions, activity draws, partner choice, service, renewal and offspring traits. The random seed of a BehaviorSpace run equals its run number, so each table is reproducible.

Collectives

Clusters are imposed (species / topics). Network modules emerge from visits; the supplied R script measures modularity but does not calculate module–cluster agreement.

Observation

Interface monitors (Appendix B) and BehaviorSpace reporters measured at the end of the run (§9.1), except E0, which records every step. All experiments write **one table per experiment**.

5. Initialisation

1. Fix world constants for the chosen `world-type` (Table 7.11).
2. Seed: `behaviorspace-run-number` in BehaviorSpace, `new-seed` in the GUI.
3. Place K cluster centres evenly on a circle of radius 12 around the world centre.
4. Resources: cluster `res-index mod K`; position = centre + N(0, `cluster-spread`) per coordinate (clamped to the world); $c_k \sim \mathrm{Beta}(2, 2)$; $q_k = 0.1$, $v_k = 0.5$, $R_k = 1$.
5. Foragers: $a_i \sim \mathrm{Beta}(\mu\kappa_a, (1-\mu)\kappa_a)$, clipped to [0.01, 0.99], with μ = `activity-mean`, κ_a = `activity-concentration`. Skill: meadow $U(0.2, 0.9)$; wiki $\mathrm{Beta}(2, 3)$, a novice-heavy group. $e_i = 0.6$; uniform effort $1/N$; zero familiarity and visit counts. Each forager starts at a random resource plus N(0, 1) noise.

6. Input data

None. The model uses no external data.

7. Submodels

7.1 Perception and movement

A forager perceives all resources within `perception-radius` r, including faded resources. If none is perceived, it turns randomly and moves one step only if the boundary permits; the opportunity is lost. Otherwise it moves directly to the selected resource and visits it within the same tick.

7.2 Partner choice

Roulette choice among perceived resources with weight

In M0–M1 the choice is neutral. More generally, the two factors are controlled independently by `adaptive-foraging?` and `feedback?`. If all candidate weights sum to zero, choice is uniform.

7.3 Trait matching (M1+)

With $d_{ik} = c_k^{\text{eff}} - s_i$, the handling efficiency is

where w = `zpd-below` and z = `zpd-above`. Whenever `trait-matching?` is off, $\eta = 1$. Too-easy visits have reduced efficiency, not zero efficiency: they can yield reward, provide service and create network links.

7.4 Consumption

Holling type II with novelty:

with $a = 2$, $h = 1$, $\kappa = 0.3$ and f_{ik} the number of earlier visits of i to k. Reward is reduced by g. In the wiki, re-reading the same text yields less. Familiarity increases after every wiki visit, including failures.

7.5 Service and feedback (M2+)

If $\eta > 0$, the forager serves the resource with probability e_i. The pollen factor π depends on the previous successful visit: π_0 if there was none, 1 if it was to the same cluster, and χ (`cross-cluster-efficiency`) otherwise. With feedback on:

$$q_k leftarrow q_k + varphi, s_i, pi, (1 - q_k), qquad v_k leftarrow v_k + varphi, pi, (1 - v_k),$$

with φ = `mutualism-strength`. A more skilled forager improves quality more. Service attempts are counted even when $\pi = 0$ or feedback is off. With feedback off, quality and vitality remain fixed and reward is restored at the end of every tick, although successive visitors can still deplete it within that tick. This is the commensal control.

7.6 Adaptive foraging (M3+)

The discrete replicator form of Valdovinos et al. (2013):

$$text{effort}_{ik} leftarrow text{effort}_{ik}, max!big(0, ; 1 + G, (g - barg_i)big),$$

The chosen entry is updated, followed by a floor of 0.001 on all entries, renormalisation to $\Sigma = 1$, and $\bar{g}_i \leftarrow 0.9\,\bar{g}_i + 0.1\,g$. G = `adaptation-rate`.

7.7 Trait development (M4)

After a visit inside the band ($-w \le d \le z$):

$$s_ileftarrows_i + lambda, nu_ik, fracd + wz + w, (1 - s_i),$$

with λ = `skill-growth`. Learning is maximal at the upper edge of the band, zero at its lower edge, and absent after failed or too-easy visits; $(1 - s_i)$ gives saturation. In the meadow this represents learned flower handling, which improves with experience and saturates (Laverty 1994; Chittka et al. 1999). In the wiki it represents learning in the ZPD; the novelty factor makes re-reading teach less. Learning requires both `trait-development?` and `trait-matching?` and uses the visit's pre-update mismatch and novelty; it does not depend on whether service occurred or reward was available.

7.8 Energy and dormancy

After each visit $e_i \leftarrow \text{clip}(e_i + \gamma g - c_0,\, 0,\, 1)$, with $\gamma = 0.5$ and c_0 = `visit-cost`. Every tick an active forager loses $b = 0.005$. Below $e_{\min} = 0.05$ it becomes dormant. A dormant forager regains 0.002 per tick and resumes at $e_{\text{wake}} = 0.3$. In the wiki, dormancy is dropout.

7.9 Resource dynamics (M2+)

Every tick:

$$R_kleftarrowR_k + rho, v_k(1 - R_k), qquadv_kleftarrow(1 - delta_v), v_k, qquadq_kleftarrow(1 - del$$

with ρ = `regeneration-rate` and $\delta_v = 0.02$ (an unattended resource fades in about 80 ticks). δ_q is 0 in the meadow and 0.003 in the wiki, where texts become outdated. Then $c_k^{\text{eff}} = c_k(1 - \sigma q_k)$ and `alive?` are updated. Fading does not remove a resource: it can still be selected, regenerate reward, receive service and recover vitality.

7.10 Renewal of faded resources (M5a)

Requires feedback (M2). Each tick, a resource with `alive? = false` is renewed with probability `renewal-rate`. The source of the newcomer is the key difference between the worlds:

Aspect	Meadow: seedling	Wiki: new article
Source	Living plant of the same cluster, chosen $\propto q$ (pollination success)	Active forager within r, chosen $\propto s$
Complexity	$c = c_{\text{parent}} + N(0, \sigma_o)$, inherited	$c = s_{\text{author}} + z + N(0, \sigma_o)$, an upper-edge target in intrinsic complexity before clipping and scaffolding
No source	No renewal in that attempt	No renewal in that attempt

σ_o = `offspring-sd`, and c is clipped to [0.01, 0.99]. The new resource starts with $q = 0.1$, $v = 0.3$, $R = 1$ and zero counts. All foragers reset their familiarity with it to 0 and their effort on it to $1/N$ (renormalised). The slot keeps its index, cluster and position, so list lengths and network dimensions stay constant. Successful-visit histories (`succ-visits`) and `prev-resource` are not reset, so the cumulative network describes resource slots across generations. The record `tick:slot:parent:author:old-c:new-c` lets R reconstruct intrinsic complexity at each visit to four-decimal precision. Visits precede renewal within a tick. Renewal is processed sequentially in random resource order; an earlier renewed resource can become a parent later in that stage.

7.11 Parameters

World constants (fixed by `world-type`; not in the interface):

Symbol	Meaning	Meadow	Wiki
w	`zpd-below`: tolerance for easier resources	0.30	0.20
z	`zpd-above`: tolerance for harder resources	0.05	0.20
χ	`cross-cluster-efficiency`	0.2	0.8
σ	`scaffolding`	0	0.5
π_0	Service factor without carried pollen	0	0.5
δ_q	Quality decay	0	0.003
ν	Novelty decay	off	$\kappa = 0.3$

Interface parameters are listed below with the experimental reference settings. The supplied file opens in the wiki world with all five switches on. Its saved slider values differ from the experimental reference in three cases: `mutualism-strength` = 0.45, `skill-growth` = 0.115 and `offspring-sd` = 0.14. BehaviorSpace explicitly sets these parameters, so the saved GUI values do not change the included experimental designs.

Parameter	Experimental reference	Level
`world-type`	meadow / wiki	All
`trait-matching?`, `feedback?`, `adaptive-foraging?`, `trait-development?`, `renewal?`	ladder switches	M1, M2, M3, M4, M5a
`num-foragers`	30	all
`num-resources`	98	all
`cluster-count`	6	all
`cluster-spread`	3	all
`perception-radius` r	6	all

Parameter	Experimental reference	Level
`activity-mean` μ	0.3	all
`activity-concentration` κ_a	2	all
`visit-cost` c_0	0.05	all
`mutualism-strength` φ	0.1	M2
`regeneration-rate` ρ	0.2	M2
`adaptation-rate` G	5	M3
`skill-growth` λ	0.02	M4
`renewal-rate`	0.05	M5a
`offspring-sd` σ_o	0.05	M5a

Internal constants: $a = 2$, $h = 1$, $\gamma = 0.5$, $\eta_{\text{low}} = 0.3$,
$b = 0.005$, $e_{\min} = 0.05$, $e_{\text{wake}} = 0.3$, recovery 0.002,
$\delta_v = 0.02$, $v_{\min} = 0.1$, $R_{\max} = 1$, effort floor 0.001,
renewed vitality 0.3, cluster ring radius 12, run length 150.

8. Model ladder and experiments

Level	Adds	Switch
M0	Heterogeneous activity, neutral choice	All five switches off
M1	Trait matching (ZPD band)	`trait-matching?`
M2	Mutualistic feedback; resource dynamics	`feedback?`
M3	Adaptive foraging	`adaptive-foraging?`
M4	Trait development (learning)	`trait-development?`
M5a	Renewal of faded resources	`renewal?`

Each level adds one mechanism, and any combination of switches is valid, so factorial designs separate the effect of each mechanism.

Experiment	Design	Runs	Question / hypotheses
E0 `E0_demo`	2 worlds, full model (M5a), measured every tick	20	Time course of one season / semester
E1 `E1_ladder`	2 worlds × 2^5 ladder switches; r = 6, λ = 0.02	1280	H1, H2, H7; contribution of each mechanism
E2 `E2_radius`	2 worlds × M2/M3 × r ∈ {3, 4.5, 6, 7.5, 9, 10.5, 12, 15, 18, 20}	800	H7r

Experiment	Design	Runs	Question / hypotheses
E3 `E3_development`	2 worlds × λ ∈ {0, 0.01, 0.02, 0.04, 0.08} × r ∈ {3, 6, 12}; M1–M4 on	600	H8
E4 `E4_renewal`	2 worlds × renewal off/on × λ ∈ {0, 0.04, 0.08} × r ∈ {3, 6}; M1–M4 on	480	H9
E5 `E5_sensitivity`	2 worlds × 3^5 factorial: mutualism {0.05, 0.1, 0.2}; regeneration {0.1, 0.2, 0.4}; adaptation {2, 5, 10}; visit cost {0.025, 0.05, 0.1}; activity mean {0.2, 0.3, 0.4}; M5a	2430	Robustness of the meadow–wiki contrasts

All experiments run 150 ticks with 20 repetitions per cell (E0: 10, E5: 5) and end-of-run measurement (E0: every tick). The random seed of a run is its run number, so the tables are reproducible; the reference checksums are listed in the README. All six experiments are embedded in the supplied `.nlogox` file.

9. Analysis pipeline in R

9.1 BehaviorSpace reporters

- **Scalars:** `share-active-foragers`, `share-alive-resources`, `mean-quality`, `mean-energy`, `mean-skill`, `skill-gain`, `fail-rate`, `service-rate`, `share-in-band`, `share-too-easy`, `cross-cluster-share`, `mean-gain-per-visit`, `unique-links`, `connectance-obs`, `effective-diet`, `visit-diet`, `renewal-count`, `share-renewed`, `mean-complexity`, `mean-complexity-alive`, `skill-complexity-gap`.
- **Strings** (records separated by `;`, fields by `:`):
  - `web-edgelist`: `forager:resource:count`;
  - `forager-traits`, `forager-skills`, `resource-traits`;
  - `visit-sequence`: `tick:forager:resource` for every successful visit;
  - `renewal-list`: `tick:slot:parent:author:old-c:new-c`.

Here, a successful visit means $\eta > 0$, including too-easy visits and visits with zero realised gain. Visit shares and rates are cumulative over attempted visits; `cross-cluster-share` counts cross-cluster services divided by services with a previous successful resource. `mean-quality` averages all resource slots. The skill–complexity gap is mean skill of active foragers minus mean effective complexity of living resources; it returns zero if either set is empty.

9.2 Metrics

The single script `MUTUS_analysis.R` reads the six BehaviorSpace tables and writes small CSV files and figures:

- checksums and cell means with SE;

- binary-network NODF (Almeida-Neto et al. 2008) and Louvain modularity (Blondel et al. 2008),
  with z-scores against curveball null models that preserve both degree sequences (Strona et al. 2014), for whole-run networks in E1–E2 and 50-tick windows in E3–E4;
- mean complexity of visited resources per window, with complexity recovered at the moment of each visit;
- the table of renewals;
- main effects of each mechanism over the E1 factorial;
- for E5, standardised regression coefficients of each output on the five varied parameters, and the sign stability of each meadow–wiki contrast across the 243 parameter cells.

Packages: `vegan` (Oksanen et al. 2022), `igraph` (Csárdi & Nepusz 2006), `dplyr`, `tidyr`, `ggplot2`.

10. Main results and limitations

10.1 Main results

1. **Trait matching structures the system; feedback makes it a mutualism (E1).** Trait matching (M1) causes 30–36 % failed visits and cuts distinct links by a third. Feedback (M2) triples resource quality but lets about half of the resources fade. Adaptive foraging (M3) halves failures (P3). Learning (M4) and renewal (M5a) mostly repair these costs.
2. **Space shapes the network; nestedness does not emerge (E1, E2).** Modularity is far above the null model at every level, including the neutral M0 ($z(Q) \approx 30$), and declines markedly as perception spans neighbouring clusters ($r = 7.5$–12). Systematic above-null nestedness is not reproduced: mean z(NODF) is generally negative, although M1 in the meadow is slightly positive (approximately 0.19). Too-hard visits fail, whereas too-easy visits retain reduced efficiency; a two-sided exclusion of links is therefore not implemented. Pattern P1 is not consistently reproduced, whereas modular structure is observed. A narrow horizon ($r = 3$) makes about half of the students drop out in the wiki, against a quarter in the meadow (H7r).
3. **Learners outgrow resources (E3).** Skill gain saturates (P4), failures halve, and the share of too-easy visits rises from 0.19 to 0.46 in the wiki (H8). Networks become more modular, not more nested, during the semester.
4. **Only the wiki follows its learners (E4).** Renewal by student authors narrows the skill–complexity gap by 0.07–0.12; renewal by seedlings makes the meadow slightly easier (H9).
5. **Robust contrasts (E5).** Across all 243 parameter cells the wiki keeps a smaller skill–complexity gap (100 % of cells), lower resource quality (91 % of cells) and a higher share of cross-cluster services (88 % of cells) than the meadow.

Differences in failure rate,
too-easy share, activity and skill gain are not robust. Retention depends mostly on `visit-cost`, quality on `mutualism-strength`.

10.2 Limitations

- **No empirical calibration.** Parameters are stylised. The wiki world is a theoretical mapping, not fitted to data from a course wiki.
- **Synchronous fading.** All resources start with vitality 0.5, so resources that are never visited fall below the threshold together at about tick 80. This shows as a step in the time course (E0). Its possible influence on end-of-run outcomes has not been isolated by an alternative-initialisation experiment.
- **Fixed populations.** There is no recruitment or departure of foragers, and no learning in the meadow beyond handling skill.
- **Renewal replaces faded resources only.** Retiring or rewriting outgrown resources is outside the scope of v1.0.
- **Sensitivity analysis is local to the reference settings.** E5 varies five parameters at three levels with 5 replicates per cell; the world constants (ZPD band, scaffolding, cross-cluster efficiency) are fixed by design.

Appendix A. Genealogy

MUTUS descends from STiMUS v2.1, which combined stigmergy (pheromone-like traces guiding target choice) and a one-way mutualism channel. MUTUS removes stigmergy and the teacher channel. It rebuilds mutualism as reciprocal, partner-choice-based interaction between two guilds, following plant–pollinator theory. In STiMUS v1, benefit-driven target choice produced preferential attachment that harmed quality. MUTUS makes this the research question: when does benefit-driven partner choice give monopoly, and when does it give coexistence?

Appendix B. Interface (v1.0)

- **Buttons:** `setup`, `go`, `go-once`.
- **Choosers and switches:** `world-type`; the five ladder switches.
- **Sliders:** the interface parameters of §7.11.
- **Monitors (9):** active foragers, alive resources, fail rate, too-easy share, mean skill, mean quality, unique links, skill–complexity gap, renewals.
- **Plots (3):**
  - `Guild state`: active foragers, alive resources, mean energy, mean quality, mean skill;
  - `Visit outcomes`: fail rate, service rate, cross-cluster share;
  - `Link accumulation (E0)`: cumulative unique links.

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